

Complex Analysis Homework 4

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15. Use the Cauchy inequalities or the maximum modulus principle to solve the following problems:

(a) Prove that if f is an entire function that satisfies

$$|z|=R|f(z)| \leq AR^k + B$$

for all $R > 0$, and for some integer $k \geq 0$ and some constants $A, B > 0$, then f is a polynomial of degree $\leq k$.

(b) Show that if f is holomorphic in the unit disc, is bounded, and converges uniformly to zero in the sector $\theta < \arg z < \varphi$ as $|z| \rightarrow 1$, then $f = 0$.

(c) Let $w_1, \dots, w_n$ be points on the unit circle in the complex plane. Prove that there exists a point z on the unit circle such that the product of the distances from z to the points w_j , $1 \leq j \leq n$, is at least 1. Conclude that there exists a point w on the unit circle such that the product of the distances from w to the points w_j , $1 \leq j \leq n$, is exactly equal to 1.

(d) Show that if the real part of an entire function f is bounded, then f is constant.

证明 (a) 由柯西不等式, $x < R$ 时

$$|f^{(n)}(x)| \leq \frac{n!}{R^n} \sup_{|\zeta|=R} |f(\zeta)| \leq n! \frac{AR^k + B}{R^n}.$$

当 $n > k$ 时, 令 $R \rightarrow \infty$, 右边趋于 0, 故 $f^{(n)} = 0$ 。因此 f 只有次数 $\leq k$ 的项, 即为次数 $\leq k$ 的多项式。

(b) 考虑辅助函数

$$g(z) = f(z)f\left(ze^{i\frac{\varphi-\theta}{2}}\right)f\left(ze^{i\frac{2(\varphi-\theta)}{2}}\right)\cdots f\left(ze^{i\frac{m(\varphi-\theta)}{2}}\right),$$

其中 m 足够大使得 $\frac{m(\varphi - \theta)}{2} > 2\pi$, 故而 g 在整个单位圆周 S^1 上均一致趋于 0, 由最大模原理有 $g \equiv 0$, 故而 f 在单位圆盘上的零点不可数, 进而 $f \equiv 0$.

(c) 考虑多项式

$$P(z) = \prod_{j=1}^n (z - w_j).$$

在单位圆周上 $|z| = 1$, 故

$$|P(z)| = \prod_{j=1}^n |z - w_j|.$$

由最大模原理, $|P(z)|$ 在闭单位圆盘上的最大值在边界取得。而

$$|P(0)| = \prod_{j=1}^n |w_j| = 1.$$

因此边界上必有一点 z 使得 $|P(z)| \geq |P(0)| = 1$ 。

函数 $\prod |z - w_j|$ 在单位圆周上连续, 而 $P(w_1) = 0$, 由介值定理, 必存在点使乘积恰好等于 1。

(d) 设 $\operatorname{Re} f(z) \leq M$ 。令

$$g(z) = e^{f(z)}.$$

则

$$|g(z)| = e^{\operatorname{Re} f(z)} \leq e^M.$$

g 是有界整函数, 由刘维尔定理, g 为常数, 从而 f 为常数。

16. Suppose f and g are holomorphic in a region containing the disc $|z| \leq 1$. Suppose that f has a simple zero at $z = 0$ and vanishes nowhere else in $|z| \leq 1$. Let

$$f_\epsilon(z) = f(z) + \epsilon g(z).$$

Show that if ϵ is sufficiently small, then

(a) $f_\epsilon(z)$ has a unique zero in $|z| \leq 1$, and

(b) if z_ϵ is this zero, the mapping $\epsilon \mapsto z_\epsilon$ is continuous.

证明 (a) 由于 f 只有 $z = 0$ 一个零点, 故而在 S^1 上非零连续, 设在 S^1 上有 $|f| \geq m > 0$, 再设在 S^1 上有 $|g| \leq M$, 故而取 $\forall \epsilon < \frac{m}{M}$, 则在 S^1 上有

$$|f| \geq m > \epsilon M \geq |\epsilon g|,$$

因此由 *Rouché* 定理, f 与 $f + \epsilon g$ 在单位圆盘内的零点个数相同.

(b) 记为 $f_\epsilon(z_\epsilon) = 0$, 设 $\epsilon_n \rightarrow \epsilon$, 我们只需证明 $z_{\epsilon_n} \rightarrow z_\epsilon$.

首先由定义, 我们有

$$\epsilon = \frac{f(z_\epsilon)}{g(z_\epsilon)}, \quad \epsilon_n = \frac{f(z_{\epsilon_n})}{g(z_{\epsilon_n})},$$

而由 $\epsilon_n \rightarrow \epsilon$, 有

$$\lim_{n \rightarrow \infty} \frac{f(z_{\epsilon_n})}{g(z_{\epsilon_n})} = \lim_{n \rightarrow \infty} \epsilon_n = \epsilon = \frac{f(z_\epsilon)}{g(z_\epsilon)}.$$

注意到 z_ϵ 是 f/g 的一阶 ϵ -点, 即 z_ϵ 是 $f/g - \epsilon$ 的一阶零点, 这是因为 $f + \epsilon g$ 的零点是唯一的. 故而在 z_ϵ 足够小的邻域上, f/g 是单射. 因此由

$$\frac{f(z_{\epsilon_n})}{g(z_{\epsilon_n})} \rightarrow \frac{f(z_\epsilon)}{g(z_\epsilon)},$$

结合 f/g 的连续性与单射性, 有 $z_{\epsilon_n} \rightarrow z_\epsilon$.

17. Let f be non-constant and holomorphic in an open set containing the closed unit disc.

(a) Show that if $|f(z)| = 1$ whenever $|z| = 1$, then the image of f contains the unit disc. [Hint: One must show that $f(z) = w_0$ has a root for every $w_0 \in \mathbb{D}$. To do this, it suffices to show that $f(z) = 0$ has a root (why?). Use the maximum modulus principle to conclude.]

(b) If $|f(z)| \geq 1$ whenever $|z| = 1$ and there exists a point $z_0 \in \mathbb{D}$ such that $|f(z_0)| < 1$, then the image of f contains the unit disc.

证明 (a) 根据 *Rouché* 定理, 对于任意的 $w_0 \in \mathbb{D}$, $f(z) - w_0$ 和 $f(z)$ 有相同的零点个数, 利用第二章练习 2 的 15 题, $f(z)$ 存在零点, 进而 $f(z) - w_0$ 存在零点, 得证

(b) 类似 (a) 只需要证明 $f(z)$ 存在零点, 不妨设零点不存在, 考虑函数 $g(z) = \frac{1}{f(z)}$, 利用最大模原理, 在单位圆盘内 $g(z) \leq 1$, 从而 $|f(z)| \geq 1$, 矛盾! 所以零点存在

19. Prove the maximum principle for harmonic functions, that is:

(a) If u is a non-constant real-valued harmonic function in a region Ω , then u cannot attain a maximum (or a minimum) in Ω .

(b) Suppose that Ω is a region with compact closure $\overline{\Omega}$. If u is harmonic in Ω and continuous in $\overline{\Omega}$, then

$$\sup_{z \in \Omega} |u(z)| \leq \sup_{z \in \overline{\Omega} - \Omega} |u(z)|.$$

[Hint: To prove the first part, assume that u attains a local maximum at z_0 . Let f be holomorphic near z_0 with $u = \Re(f)$, and show that f is not open. The second part follows directly from the first.]

证明 (a) 反证, 不妨设 u 在 z_0 处取得极大值, 根据 u 的调和性质于是可以取 f 在 z_0 附近时全纯的而且 $u = \Re(f)$, 但由于 z_0 处 f 是实部极大, 所以 f 不是开的, 进而矛盾. 得证

(b) 紧集上连续函数一定有最值, 从而根据前一问自然有

$$\sup_{z \in \Omega} |u(z)| \leq \sup_{z \in \overline{\Omega} - \Omega} |u(z)|.$$

21. Certain sets have geometric properties that guarantee they are simply connected.

(a) An open set $\Omega \subset \mathbb{C}$ is convex if for any two points in Ω , the straight line segment between them is contained in Ω . Prove that a convex open set is simply connected.

(b) More generally, an open set $\Omega \subset \mathbb{C}$ is star-shaped if there exists a point $z_0 \in \Omega$ such that for any $z \in \Omega$, the straight line segment between z and z_0 is contained in Ω . Prove that a star-shaped open set is simply connected. Conclude that the slit plane $\mathbb{C} - \{(-\infty, 0]\}$ (and more generally any sector, convex or not) is simply connected.

(c) What are other examples of open sets that are simply connected?

证明 (a) 若凸集 Ω 不是单连通, 则一定存在闭曲线 γ 不同伦于一个点, 因此 γ 内部有个点不属于 Ω , 而 γ 是环绕着这个点的, 因此能在 γ 上取两个点, 与该点三点共线, 而 $\gamma \subset \Omega$, 因此与凸集矛盾.

(b) 对于 Ω 内的任意闭合曲线 $p(s)$, 我们作

$$H(r, s) = rz_0 + (1 - r)p(s), \quad r \in [0, 1].$$

由于 $p(s) \subset \Omega$ 并且 Ω 是关于 z_0 的星状区域, 故而 $H(r, s) \in \Omega, \forall r, s$, 并且易知 H 连续, 而 $H(0, s) = p(s), H(1, s) = z_0$, 即 $p(s)$ 同伦于 z_0 点, 注意到 $p(s)$ 是任意的闭曲线, 故而所有闭曲线均同伦于点, 因此 Ω 单连通.

(c) 整个复平面

22. Show that there is no holomorphic function f in the unit disc $\mathbb{D}$ that extends continuously to $\partial\mathbb{D}$ such that $f(z) = 1/z$ for $z \in \partial\mathbb{D}$.

证明 不妨设 f 存在, 考虑 $g(z) = zf(z) - 1$, 在边界上恒为 0, 从而根据最大模原理得到 $g(z) = 0 \quad z \in \partial\mathbb{D}$, 代入 $z = 0$, 有 $1 = 0$, 矛盾!

1. Consider a holomorphic map on the unit disc $f : \mathbb{D} \rightarrow \mathbb{C}$ which satisfies $f(0) = 0$. By the open mapping theorem, the image $f(\mathbb{D})$ contains a small disc centered at the origin. We then ask: does there exist $r > 0$ such that for all $f : \mathbb{D} \rightarrow \mathbb{C}$ with $f(0) = 0$, we have $D_r(0) \subset f(\mathbb{D})$?

(a) Show that with no further restrictions on f , no such r exists. It suffices to find a sequence of functions $\{f_n\}$ holomorphic in $\mathbb{D}$ such that $1/n \notin f(\mathbb{D})$. Compute $f'_n(0)$, and discuss.

(b) Assume in addition that f also satisfies $f'(0) = 1$. Show that despite this new assumption, there exists no $r > 0$ satisfying the desired condition. [Hint: Try $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$.]

The Koebe-Bieberbach theorem states that if in addition to $f(0) = 0$ and $f'(0) = 1$ we also assume that f is injective, then such an r exists and the best possible value is $r = 1/4$.

(c) As a first step, show that if $h(z) = \frac{1}{z} + c_0 + c_1z + c_2z^2 + \cdots$ is analytic and injective for $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$. [Hint: Calculate the area of the complement of $h(D_\rho(0) - \{0\})$ where $0 < \rho < 1$, and let $\rho \rightarrow 1$.]

(d) If $f(z) = z + a_2z^2 + \cdots$ satisfies the hypotheses of the theorem, show that there exists another function g satisfying the hypotheses of the theorem such that $g^2(z) = f(z^2)$. [Hint: $f(z)/z$ is nowhere vanishing so there exists ψ such that $\psi^2(z) = f(z)/z$ and $\psi(0) = 1$. Check that $g(z) = z\psi(z^2)$ is injective.]

(e) With the notation of the previous part, show that $|a_2| \leq 2$, and that equality holds if and only if

$$f(z) = \frac{z}{(1 - e^{i\theta}z)^2} \quad \text{for some } \theta \in \mathbb{C}.$$

[Hint: What is the power series expansion of $1/g(z)$? Use part (c).]

(f) If $h(z) = \frac{1}{z} + c_0 + c_1z + c_2z^2 + \cdots$ is injective on $\mathbb{D}$ and avoids the values z_1 and z_2 , show that $|z_1 - z_2| \leq 4$. [Hint: Look at the second coefficient in the power series expansion of $1/(h(z) - z_j)$.]

(g) Complete the proof of the theorem. [Hint: If f avoids w , then $1/f$ avoids 0 and $1/w$.]

证明 (a) 取 $f_n = \frac{z}{n}$, 则 $f'_n(0) = \frac{1}{n}$

(b) 取 $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$, 则对于 $-\epsilon$, 无法取到

(c) 首先注意到 c_0 与本题条件和结论均无关, 因此不妨设 $c_0 = 0$. 再注意到, 如果用 $\lambda h(\lambda z)$ 代替 $h(z)$, 其中 $|\lambda| = 1$, 本题条件结论不变, 故而不妨再设 $c_1 \in \mathbb{R}$.

我们先设

$$U_r = \{z : |z| < r\}, \quad C_r = \{z : |z| = r\}, \quad V_r = \{z : r < |z| < 1\},$$

其中 $r \in (0, 1)$. 由开映射定理, $h(U_r)$ 是 ∞ 的某个邻域, 即 $\mathbb{C} \setminus h(U_r)$ 是 $\mathbb{C}$ 上的紧集, 再由 h 是单射知 $h(U_r), h(C_r), h(V_r)$ 互不相交. 我们令

$$h(z) = \frac{1}{z} + c_1 z + \varphi(z), \quad z \in \mathbb{D},$$

再设 $h = u + iv$, 作 $z = re^{i\theta}$, 则有

$$u = A \cos \theta + \operatorname{Re} \varphi, \quad v = -B \sin \theta + \operatorname{Im} \varphi,$$

其中

$$A = \frac{1}{r} + c_1 r, \quad B = \frac{1}{r} - c_1 r.$$

于是有

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1 + \frac{2 \cos \theta}{A} \operatorname{Re} \varphi + \left(\frac{\operatorname{Re} \varphi}{A} \right)^2 - \frac{2 \sin \theta}{B} \operatorname{Im} \varphi + \left(\frac{\operatorname{Im} \varphi}{B} \right)^2.$$

注意到 φ 在 $z = 0$ 处至少是二阶零点, 即

$$\varphi = O(r^2), \quad r \rightarrow 0,$$

而由 A, B 表达式知

$$\frac{1}{A} = O(r), \quad \frac{1}{B} = O(r), \quad r \rightarrow 0,$$

故而有

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1 + O(r^3), \quad r \rightarrow 0.$$

因此, $\exists \eta > 0$ 使得对于任意充分小的 r 均有

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} < 1 + \eta r^3,$$

其中 $z = re^{i\theta}$. 也就是说, $h(C_r)$ 的像包含在椭圆

$$E_r : \frac{u^2}{A^2} + \frac{v^2}{B^2} = 1 + \eta r^3$$

内部, 进而有 $E_r \subset h(U_r)$, $h(V_r)$ 也在 E_r 内部.

一方面, 椭圆 E_r 所围面积为

$$\pi AB(1 + \eta r^3) = \pi \left(\frac{1}{r} + c_1 r \right) \left(\frac{1}{r} - c_1 r \right) (1 + \eta r^3) \leq \frac{\pi}{r^2} (1 + \eta r^3).$$

另一方面, $h(V_r)$ 的面积为

$$\begin{aligned} |h(V_r)| &= \iint_{V_r} |h'(z)|^2 dx dy \\ &= \int_r^1 dt \int_0^{2\pi} \left(-\frac{1}{t^2} e^{-2i\theta} + \sum_{k=1}^{\infty} c_k t^{k-1} e^{i(k-1)\theta} k \right) \left(-\frac{1}{t^2} e^{2i\theta} + \sum_{k=1}^{\infty} \overline{c_k} t^{k-1} e^{-i(k-1)\theta} k \right) t d\theta \\ &= 2\pi \int_r^1 \left(\frac{1}{t^3} + \sum_{k=1}^{\infty} k^2 |c_k|^2 t^{2k-1} \right) dt \\ &= \pi \left(\frac{1}{r^2} - 1 + \sum_{k=1}^{\infty} k |c_k|^2 (1 - r^{2k}) \right). \end{aligned}$$

于是乎, 我们有

$$\frac{\pi}{r^2} (1 + \eta r^3) \geq \pi \left(\frac{1}{r^2} - 1 + \sum_{k=1}^{\infty} k |c_k|^2 (1 - r^{2k}) \right),$$

化简有

$$\sum_{n=1}^{\infty} n |c_n|^2 (1 - r^{2n}) \leq 1 + \eta r,$$

由 $0 < r < 1$ 的任意性, 令 $r \rightarrow 0$ 即得

(d) 由于 $f(z)/z$ 没有零点, 所以存在 ψ 使得 $\psi^2(z) = f(z)/z$, 设

$$g(z) = z\psi(z^2)$$

计算得 $g(0) = 0, g'(0) = 1$

如果 $g(z) = g(w)$, 则自然有 $f(z^2) = f(w^2)$, 若 $z \neq w$, 则 $z = -w$, 从而

$$g(z) = z\psi(z^2) = -w\psi^2(w^2) = -g(w)$$

, 进而 $g(z) = g(w) = 0$, 从而 $f(z) = f(w) = 0$, 于是 $z = w$, 综上 g 是单射

(e) 计算得

$$g(z) = z(1 + \frac{1}{2}a_2z^2 + \cdots)$$

设 $h(z) = \frac{1}{g(z)}$, 于是

$$h(z) = \frac{1}{z}(1 - \frac{1}{2}a_2z^2 + \cdots)$$

根据 g 的性质可得 h 满足 c 的条件, 利用 c 有

$$\frac{1}{2}|a_2| \leq 1$$

即 $|a_2| \leq 2$, 等号成立仅当 $h(z) = \frac{1}{z}(1 - \frac{1}{2}a_2z^2)$ 且 $a_2 = 2e^{i\theta}$ 于是 $g(z) = \frac{z}{1 - e^{i\theta}z^2}$, 从而

$$f(z) = \frac{z}{(1 - e^{i\theta}z)^2} \quad \text{for some } \theta \in \mathbb{C}.$$

(f) 考虑函数 $\frac{1}{h(z) - z_j}$, 其中 $j = 1, 2$. 由条件 $\frac{1}{h(z) - z_j}$ 全纯, 并且有

$$\frac{1}{h(z) - z_j} = z - (c_0 - z_j)z^2 + \cdots.$$

于是

$$\frac{1}{h(0) - z_j} = 0, \quad \left(\frac{1}{h(z) - z_j} \right)' \Big|_{z=0} = 1.$$

由 (e) 的结论, 有

$$|c_0 - z_j| \leq 2, \quad j = 1, 2,$$

故而有

$$|z_1 - z_2| \leq |c_0 - z_1| + |c_0 - z_2| \leq 4.$$

(g) 设单射全纯函数 $f(z) = z + a_2 z^2 + \cdots$ 是满足定理条件的函数, 设 $w \in \mathbb{C}$ 但 $w \notin f(\mathbb{D})$, 则考虑函数 $1/f$, 该函数满足 (f), (c) 中 h 的条件, 且取不到 $0, \frac{1}{w}$, 故而由 (f) 的结论可知

$$\left| \frac{1}{w} \right| \leq 4,$$

即有

$$|w| \geq \frac{1}{4},$$

因此有 $B(0, \frac{1}{4}) \subset f(\mathbb{D})$, 定理得证.